



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

LAB REPORTS

Course no: **MTE 3206**

Course Title: **Hydraulic and Pneumatic Control Sessional**

Experiment No: **03**

Name of the Experiment: Simulation and Operation of pneumatic circuits for Latching, timers, logic sequencing and counter operations.

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Date of Submission: April 29, 2025

Experiment No.: 03

Experiment Name:

Simulation and Operation of Pneumatic Circuits for Latching, Timers (On Delay & Off Delay), Logic Sequencing ($A+B+A-B-$ and $A+A-B+B-$), and Counter Operations using FluidSIM.

Objectives

- To simulate and analyze latching circuits using FluidSIM, exploring their functionality and applications in pneumatic systems.
- To study and understand the working principles of On Delay and Off Delay pneumatic timers, and their impact on system operation.
- To implement and observe the behavior of $A+B+A-B-$ and $A+A-B+B-$ logic sequencing circuits in pneumatic systems, demonstrating their sequence control.
- To demonstrate and investigate counter operations within pneumatic systems, highlighting their use in various automated processes.

Theory

In pneumatics, various components are used to control the sequence and timing of actuator operations.

Latching Circuit: Latching was used to maintain the actuator's state even after the control signal was removed. A self-holding circuit was built using a push-button and a directional control valve.

On Delay Timer (Normally Closed Time Delay Valve): An On Delay Timer was used to delay the extension of an actuator after a signal was given. A normally closed (NC) time delay valve was used for this purpose. The valve only allowed air through after a preset time [1].

Off Delay Timer (Normally Open Time Delay Valve): An Off Delay Timer delayed the retraction of an actuator after the signal was removed. A normally open (NO) time delay valve was used, which closed after a set time, allowing the actuator to remain extended momentarily [1].

$A+B+A-B-$ Sequencing: This sequencing involved extending cylinder A, followed by cylinder B. Then cylinder A retracted, followed by the retraction of cylinder B. It was achieved using roller lever valves and logical sequencing.

$A+A-B+B-$ Sequencing: This sequence required cylinder A to extend and retract first, then cylinder B to extend and retract. This ensured the complete action of one actuator before the other started.

Counter Operations: Counters were used to control the number of cycles an actuator performed. When the pre-set count was reached, the operation was halted automatically.

Required Apparatus

- FluidSIM Software
- Single-acting & Double-acting pneumatic cylinders
- 3/2 & 4/2-way directional control valves (DCVs)
- Time delay valves (normally closed and normally open)
- Pneumatic push buttons
- Limit (roller lever) valves
- Pneumatic counter
- Air source (simulated in software)
- Connecting hoses (simulated)

Operations

Latching Circuit: A latching mechanism was simulated using a 5/2 Directional Control Valve (DCV). A push button was used to energize the valve, initiating the operation. The output was latched using a self-holding path, ensuring that once the button was released, the actuator remained extended. This self-holding feature allowed the actuator to stay in the extended position without continuous input, effectively demonstrating a latching circuit in the pneumatic system [2].

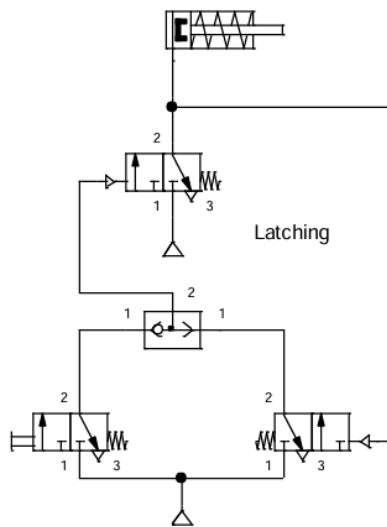


Fig.1. Latching Operation

On Delay Timer Circuit (Normally Closed): In this setup, a Normally Closed (NC) time delay valve was connected before the Directional Control Valve (DCV) that controls the cylinder. When the push button was pressed, air flowed to the timer. After a preset delay, the timer opened, allowing air to reach the DCV and extending the cylinder. This circuit demonstrated the working principle of an On Delay Timer, where the output action (cylinder extension) occurs only after the

specified delay time.

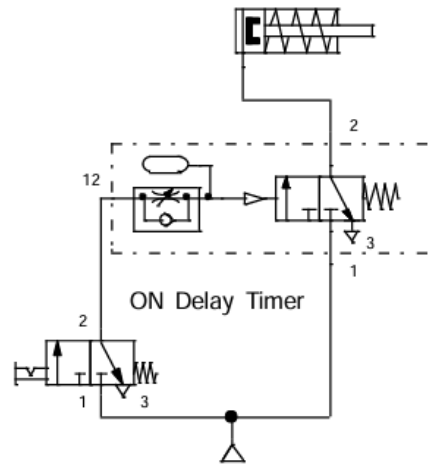


Fig.2. ON Delay Timer Operation

Off Delay Timer Circuit (Normally Open): In this configuration, a Normally Open (NO) time delay valve was used. When the button was pressed, the air flowed through the timer. Upon releasing the button, the air continued to flow momentarily through the timer before being cut off. This allowed the actuator to remain extended for a short duration after the button was released, demonstrating the working principle of an Off Delay Timer. The actuator's extension was maintained for a brief period, even after the input signal was removed, showcasing the delay in output.

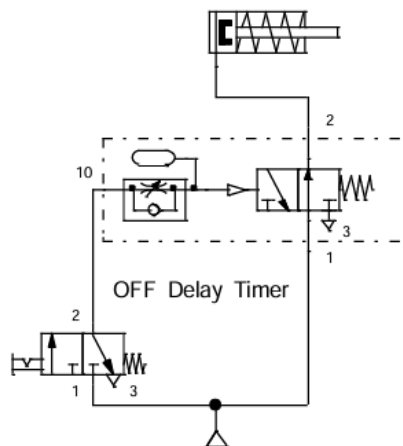


Fig.3. OFF Delay Timer Operation

A+B+A-B- Logic Sequencing: In this operation, two double-acting cylinders (A and B) were used to demonstrate the A+B+A-B- logic sequencing. Roller lever valves, positioned at the ends of the cylinders, were employed to detect the positions of the cylinders. Proper sequencing was achieved using signal lines and logic control, ensuring the correct order of operations. The sequence followed was A+ (cylinder A extended), B+ (cylinder B extended), A- (cylinder A retracted), and B- (cylinder B retracted), successfully completing the logic sequence as intended.

This demonstrated the integration of logic control in pneumatic systems for coordinated motion.

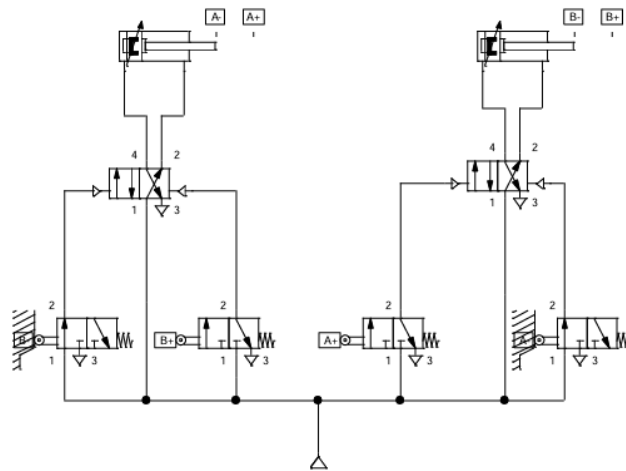


Fig.4. A+B+A-B- Logic Sequencing Operation

A+A-B+B- Logic Sequencing: In this operation, Cylinder A completed a full cycle (extension and retraction) before Cylinder B started its cycle. Signal dependency was implemented using limit valves to ensure that Cylinder B's sequence would only begin once Cylinder A had fully returned to its initial position. This sequencing logic prevented both cylinders from operating simultaneously and ensured that Cylinder B's motion was triggered only after Cylinder A had retracted, demonstrating controlled and dependent sequencing in pneumatic systems [2].

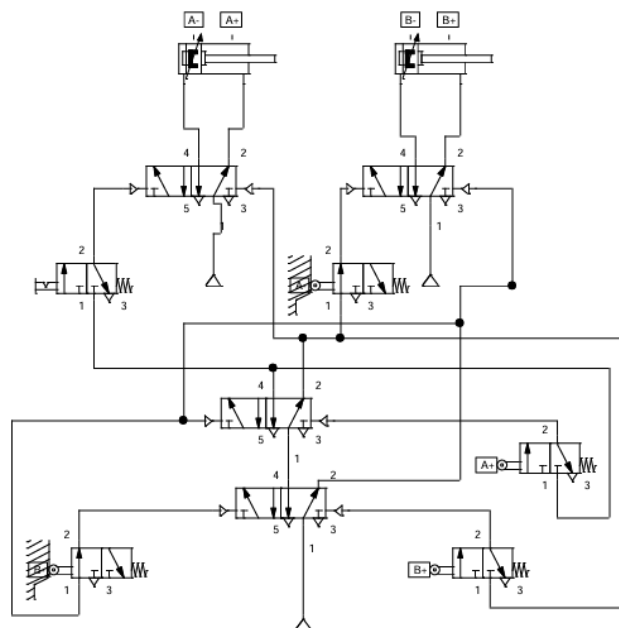


Fig.5. A+A-B+B- Logic Sequencing

Counter Operation: In this setup, a pneumatic counter was incorporated into the circuit to track the movement of the cylinder. The counter monitored the number of full cycles the cylinder completed, incrementing with each cycle. Once the preset number of operations was reached, the actuator stopped automatically. This demonstrated the use of a pneumatic counter to control the number of cycles in a pneumatic system, ensuring that the actuator ceased movement after completing the desired number of operations.

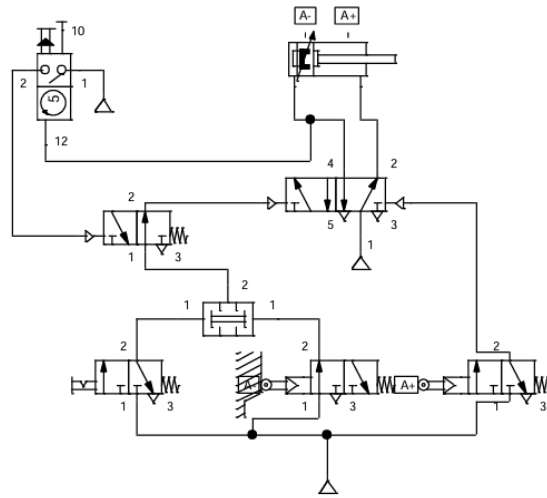


Fig.6. Counter Operation

Discussion & Conclusion

The circuits were successfully simulated in FluidSIM, and each operation was thoroughly verified. The latching circuit effectively demonstrated the memory function in pneumatic systems, allowing the actuator to remain in its last position without continuous input. The On and Off Delay timers were crucial in providing necessary time delays, ensuring safety and enhancing process accuracy in the system. The logic sequencing circuits ($A+B+A-B-$ and $A+A-B+B-$) ensured structured and coordinated operations of multiple actuators, highlighting their widespread use in industrial automation for maintaining precise control over movements.

The counter circuits demonstrated their application in batch operations, where repetitive tasks need to be limited to a set number of cycles. The integration of sensors, such as roller lever valves, enabled real-time feedback and facilitated logic-based decision-making for accurate control. Overall, the operations underscored the critical role of timing, sequencing, and feedback in the design and functioning of automated pneumatic systems, reinforcing their significance in modern industrial applications.

References

- [1] Festo Didactic, FluidSIM Pneumatics – User Manual, Festo AG & Co. KG, 2020.
- [2] J. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, 4th

ed., Boston, MA: Pearson, 2015.